Experiment No:5
Data Collection and Initial Exploration

AIM:

To collect, load, and perform initial exploration of the diabetes dataset.

Materials Needed:

Diabetes dataset (available from sources such as UCI Machine Learning Repository or Kaggle)

Python with libraries: pandas, numpy, matplotlib, seaborn

CODE:

```
  import pandas as pd
db=pd.read_csv("/content/diabetes_prediction_dataset.csv")
print(db.head())
print(db.info())
print(db.describe())
import matplotlib.pyplot as plt
import seaborn as sns
db.hist(bins=50,figsize=(20,15))
plt.show()
sns.pairplot(db)
plt.show()
```

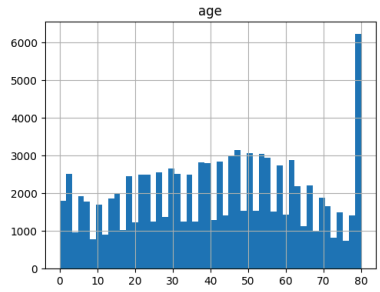
OUTPUT:

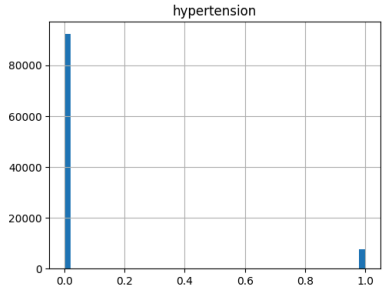
```
  gender   age  hypertension  heart_disease smoking_history    bmi  \
0  Female  80.0           0              1           never  25.19
1  Female  54.0           0              0         No Info  27.32
2    Male  28.0           0              0           never  27.32
3  Female  36.0           0              0         current  23.45
4    Male  76.0           1              1         current  20.14

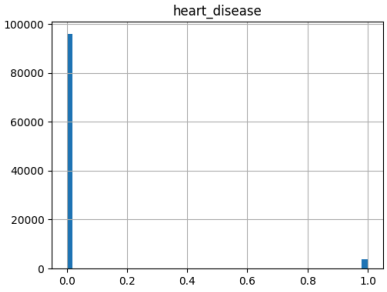
   HbA1c_level  blood_glucose_level  diabetes
0          6.6                  140         0
1          6.6                   80         0
2          5.7                  158         0
3          5.0                  155         0
4          4.8                  155         0
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 9 columns):
 #   Column               Non-Null Count   Dtype
```

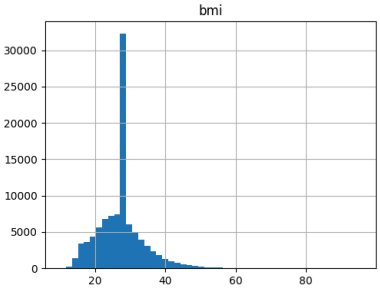
```
---  ------               --------------   -----
 0   gender               100000 non-null  object
 1   age                  100000 non-null  float64
 2   hypertension         100000 non-null  int64
 3   heart_disease        100000 non-null  int64
 4   smoking_history      100000 non-null  object
 5   bmi                  100000 non-null  float64
 6   HbA1c_level          100000 non-null  float64
 7   blood_glucose_level  100000 non-null  int64
 8   diabetes             100000 non-null  int64
dtypes: float64(3), int64(4), object(2)
memory usage: 6.9+ MB
None
                age  hypertension  heart_disease            bmi  \
count  100000.000000  100000.00000  100000.000000  100000.000000
mean       41.885856       0.07485       0.039420      27.320767
std        22.516840       0.26315       0.194593       6.636783
min         0.080000       0.00000       0.000000      10.010000
25%        24.000000       0.00000       0.000000      23.630000
50%        43.000000       0.00000       0.000000      27.320000
75%        60.000000       0.00000       0.000000      29.580000
max        80.000000       1.00000       1.000000      95.690000

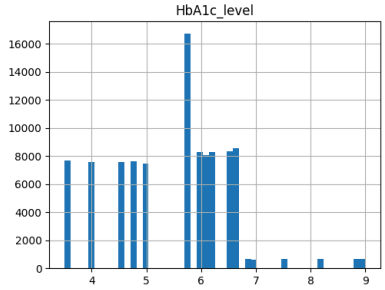
         HbA1c_level  blood_glucose_level       diabetes
count  100000.000000        100000.000000  100000.000000
mean        5.527507           138.058060       0.085000
std         1.070672            40.708136       0.278883
min         3.500000            80.000000       0.000000
25%         4.800000           100.000000       0.000000
50%         5.800000           140.000000       0.000000
75%         6.200000           159.000000       0.000000
max         9.000000           300.000000       1.000000
```

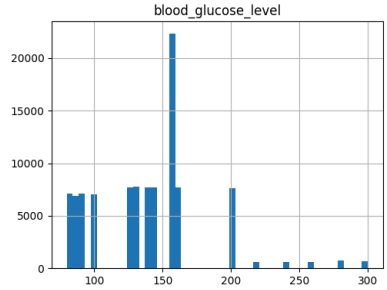

age
0
1000
2000
3000
4000
5000
6000
0
10
20
30
40
50
60
70
80

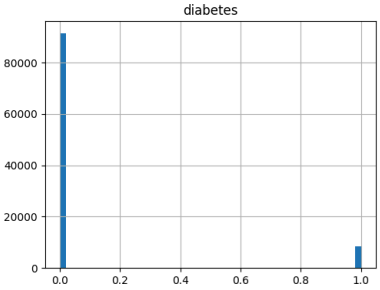

hypertension
0
20000
40000
60000
80000
0.0
0.2
0.4
0.6
0.8
1.0

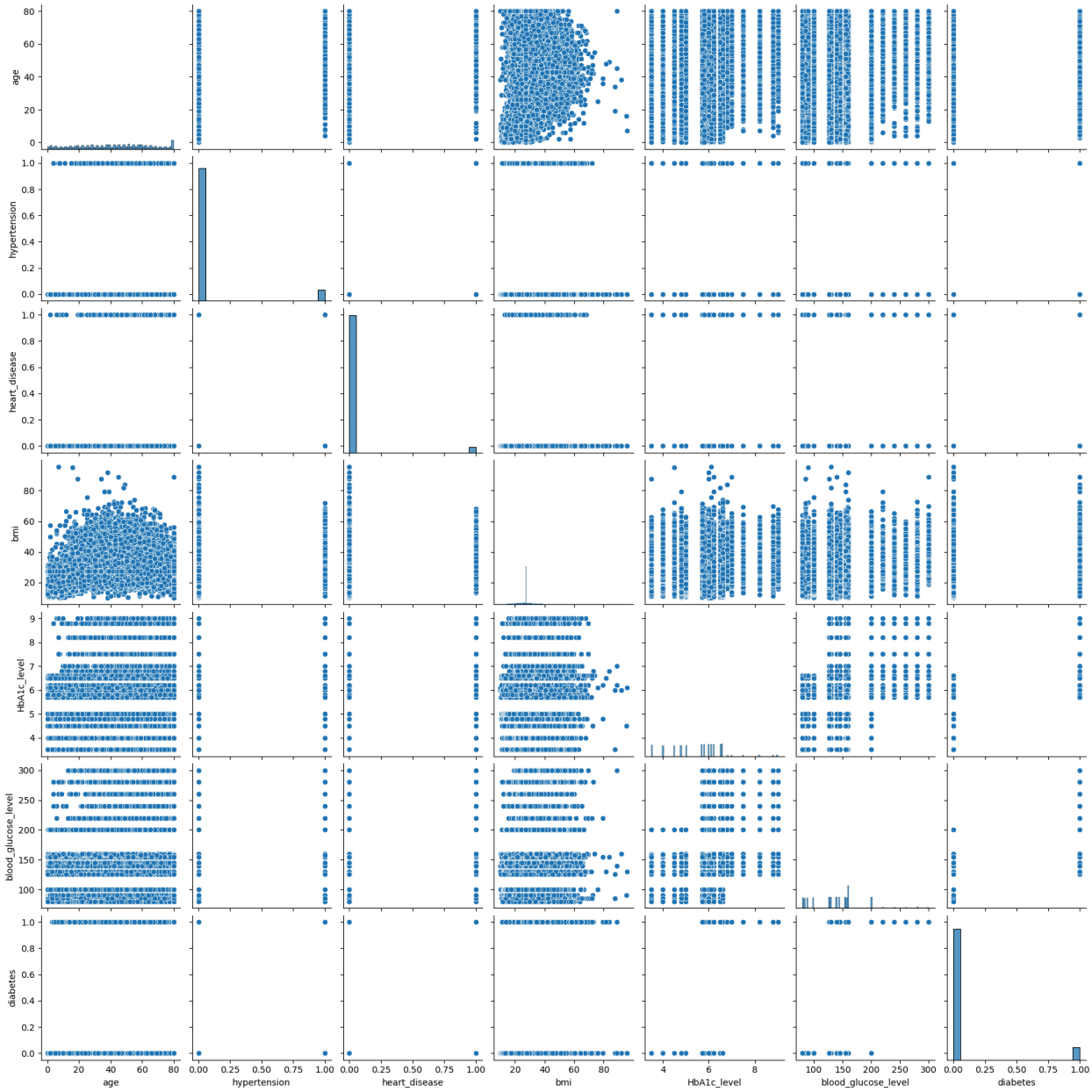

heart_disease
0
20000
40000
60000
80000
100000
0.0
0.2
0.4
0.6
0.8
1.0


bmi
0
5000
10000
15000
20000
25000
30000
20
40
60
80


HbA1c_level
0
2000
4000
6000
8000
10000
12000
14000
16000
4
5
6
7
8
9


blood_glucose_level
0
5000
10000
15000
20000
100
150
200
250
300


diabetes
0
20000
40000
60000
80000
0.0
0.2
0.4
0.6
0.8
1.0



RESULT:

The diabetes dataset was successfully loaded and initially explored.the first few records,dataset,structured and summary statistics were displayed histogram showed the distribution of each features and the pair plot showed the relationship between the variables.